

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
(An Autonomous Institution affiliated to VTU, Belagavi)  
**Fifth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**  
January 2023

**20BT504 / 17BT504 – GENETIC ENGINEERING**

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.  
2) Draw diagrams wherever necessary.

| Unit – I   |                                                                                                                                            | Marks | BT* | CO* | PO*   |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-------|
| 1. a)      | With appropriately labeled diagram of an example, describe the general characters of a plasmid.                                            | 10    | L*1 | 1   | 1.2.1 |
| b)         | Write short note on: i) PUC series vectors; ii) Phage vectors.                                                                             | 5+5   | L1  | 1   | 1.2.1 |
| 2. a)      | Brief on: i) Cosmids; ii) Promoters.                                                                                                       | 5+5   | L1  | 1   | 1.2.1 |
| b)         | Discuss in detail about the various methods of screening recombinant bacteria and phages with respective principles involved.              | 10    | L1  | 1   | 1.2.1 |
| Unit – II  |                                                                                                                                            |       |     |     |       |
| 3. a)      | Write short note on: i) Terminal transferases; ii) Ligases.                                                                                | 5+5   | L1  | 2   | 1.2.1 |
| b)         | With respect to Restriction enzymes answer the following:<br>i) Definition; ii) Nomenclature; iii) Classification with examples.           | 10    | L1  | 2   | 1.2.1 |
| 4. a)      | Write short note on: i) Alkaline phosphatases; ii) Exonucleases.                                                                           | 5+5   | L1  | 2   | 1.2.1 |
| b)         | Give a detailed account of Nucleases and Nuclease inhibitors. Add a note on their significance on rDNA technology                          | 10    | L1  | 2   | 1.2.1 |
| Unit – III |                                                                                                                                            |       |     |     |       |
| 5. a)      | With neat labelled diagrams, discuss the construction, screening of cDNA library.                                                          | 10    | L1  | 3   | 1.2.1 |
| b)         | Write short note on: i) cDNA Library vs Genomic DNA library; ii) Non-radioactive probes.                                                   | 5+5   | L1  | 3   | 1.2.1 |
| 6. a)      | With appropriate illustrations, discuss the principle of Polymerase Chain Reaction. Add a note on its significance in Biomedical Research. | 10    | L1  | 3   | 1.2.1 |
| b)         | Write short note on: i) Nucleic acid detection; ii) Plasmid isolation.                                                                     | 5+5   | L1  | 3   | 1.2.1 |
| Unit – IV  |                                                                                                                                            |       |     |     |       |
| 7. a)      | Give a detailed account on physical and chemical methods employed for gene transfer.                                                       | 10    | L1  | 4   | 2.1.2 |
| b)         | Write short note on: i) Edible vaccines; ii) Transgenic mice.                                                                              | 5+5   | L1  | 4   | 2.1.2 |
| 8. a)      | Give a detailed account of Ti plasmids and its role in plant biotechnology.                                                                | 10    | L1  | 4   | 2.1.2 |
| b)         | Brief on: i) Ri Plasmids; ii) Stem cells.                                                                                                  | 5+5   | L1  | 4   | 2.1.2 |
| Unit – V   |                                                                                                                                            |       |     |     |       |
| 9. a)      | Define Gene therapy. Discuss in detail about the approaches for gene therapy.                                                              | 10    | L1  | 5   | 2.1.2 |
| b)         | Comment with suitable example, elaborate Germline gene therapy.                                                                            | 10    | L1  | 5   | 2.1.2 |
| 10.        | Discuss in detail about Gene therapies for Cancer – Strategies, challenges and success.                                                    | 20    | L1  | 5   | 2.1.2 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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